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A Novel Approach for Bluetooth Mesh Network Detection using Machine Learning and Correlation Analysis

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Abstract

Bluetooth (BT) mesh networks are used in several ad hoc network communication scenarios, including natural disasters, calamities, war zones, etc. However, there are situations where Bluetooth mesh networks are misused by anti-social elements during riots etc. As of now there no such research article that showcases detection of BT mesh networks using machine learning tools. Therefore, there is a need to build novel methods for detecting the very presence of Bluetooth mesh networks to aid the law enforcement agencies. This paper contributes to building a new dataset for the aid of machine learning tools to detect the presence of Bluetooth mesh networks. The dataset is extracted from a simulated Bluetooth mesh network with 52 BT network traffic attributes, using the Bridgefy application. Further, the dataset is optimized by identifying prime features, using label encoding, one-hot encoding, correlation analysis and evaluated by various Machine Learning (ML) tools. An accuracy of 99.6% is achieved after optimizing the feature set.

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Keywords: Bluetooth Security, Bluetooth Mesh Networks, Machine Learning, Adhoc Networks, Feature reduction, Correlation

1. Introduction

Mesh networking apps are a type of mobile app that allows users to communicate with each other directly, without the need for a central server or internet connection [1]. They work by creating a mesh network, which is a decentralized network of devices that are interconnected with each other's [2] [3]. The reason behind the usage of mesh networking apps is to provide communication in natural disasters but the utilization of such apps is in negative track. Few examples include, like helping to terrorists through building their communication between them and

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making their work easier, these is main drawback of usage of such apps [4]. Mesh networking apps have a few advantages over traditional communication apps [5]. For example, they are more resilient to disruptions, as they can continue to operate even if one or more devices in the network are unavailable [6]. They are also more secure, as they do not rely on a central server that could be hacked [7].

Now on disadvantages where Mesh networking can drain the battery of devices more quickly than traditional networks [8]. While mesh networking can be more secure than traditional networks, there are still some security risks to be aware of [9]. For example, attackers could potentially try to infiltrate the network and gain access to user data. Mesh networking may not be compatible with all devices. It is important to check the compatibility requirements before using a mesh network. In the year 2019, Citizenship Amendment Act (CAA) protests were happened which are a controversial law [10]. During the protests, the Indian government-imposed internet shutdowns in several areas. This made it difficult for protesters to communicate with each other. The bridgefy app, which works using Bluetooth, allowed protesters to communicate even without internet access [11]. The next reports to appear centred on the Delhi riots, that occurred in 2020. Here the rise in downloads was attributed to an Internet shutdown occurring during the same period. In 2019, inspired by Hong Kong protesters' use of the Bridgefy app, organizers of a street protest downloaded the app. There is an urge to research on techniques to detect the very presence of the mesh networks.

In this paper, we propose a new approach to mesh networking app detection that uses packet sniffing and analysis through ML. Packet sniffing is the process of capturing network traffic packets [12]. Packet analysis is the process of examining network traffic packets to extract information about the network traffic [13]. We propose to train a ML model to identify mesh networking apps [14] by analysing the features of the packets that they generate. The features that we extract from the packet analysis may include things like packet size, packet header information, packet payload data and others. We train the ML model on a dataset of packets from mesh networking apps, non-mesh networking apps and smart devices. Once the ML model is trained, we can use it to detect mesh networking apps in real time by capturing and analysing network traffic packets [15].

2. Methodology

2.1 Mesh formation using bridgefy:

The flow chart of the proposed system to detect the very presence of BT mesh network is shown in Fig. 1. Firstly, a BT mesh network is simulated utilizing the Bridgefy application, wherein each mobile device seamlessly assumes the role of a network node, collectively forming a sophisticated and interconnected network fabric.

2.2 Capturing using nRF52840 dongle:

This step is dedicated to the utilization of the nRF52840 dongle for capturing critical data of the Bluetooth mesh network. The purpose is to capture, analyse, and record the intricate details of network communication to gain insights into the network's behaviour.

2.3 Feature extraction:

Feature extraction plays an indispensable role by facilitating the conversion of raw data collected during Bluetooth mesh network operation into a desired format. This desired format serves as a foundation for efficient analysis more critically, for the identification of nuanced network behaviours. The extracted features, meticulously distilled from the raw data, function as the elemental components underpinning subsequent analytical and detection methodologies.

2.4 *Feature selection*:

Feature selection is a precise process that involves choosing the most relevant and informative attributes from a pool of available features. This step is critical for streamlining the data and ensuring that only the most significant and discriminative characteristics are retained for analysis. By carefully selecting the appropriate features, we can improve the efficiency of our analytical models, reduce computational complexity, and enhance the interpretability of their results. It enables us to focus on the key attributes that are most pertinent to identifying and characterizing Bluetooth mesh network apps like Bridgefy, enhancing the precision and effectiveness of our detection methods.

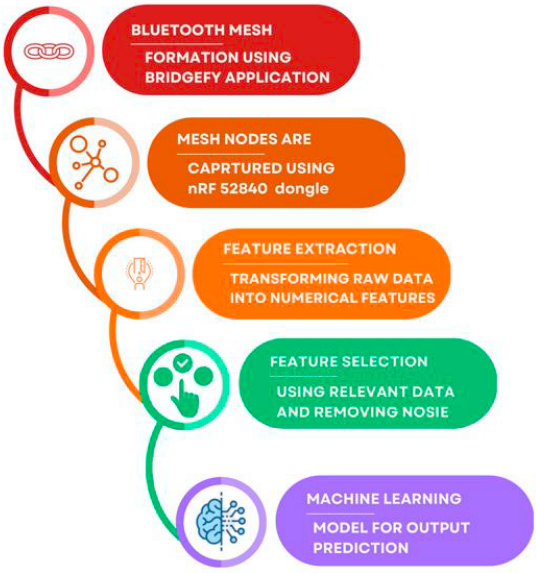


Fig. 1. Flow chart of the proposed system

2.5 *Machine learning*:

A machine learning model serves as the analytical powerhouse of the entire process, with the capacity to autonomously detect, classify, and interpret intricate patterns within data. By leveraging algorithms and data-driven techniques, the models transform extracted features into actionable insights, enabling precise identification and characterization of Bluetooth mesh network apps like Bridgefy. These models play a crucial role in automating the detection process and can uncover patterns in network behaviours, ultimately enhancing the accuracy and efficiency.

3. Dataset Collection

3.1 Feature extraction

The goal is to construct a dataset for Machine Learning models to detect Mesh and non-mesh BL traffic. Firstly, BLE mesh traffic is simulated using Bridgfy mobile application and earmarked as BLE Mesh traffic. Later, non-mesh BL traffic is simulated from various BLE devices such as smartwatches, Bluetooth neckbands, and similar consumer electronics.

To capture the network activities, an NRF52840 dongle was employed to sniff the Bluetooth traffic, while Wireshark served as the monitoring tool. The NRF52840 dongle was configured to intercept BLE communication, enabling the acquisition of raw data packets. These raw data packets were saved in a PCAP file format. To facilitate further analysis, the PCAP files were transformed into a structured CSV format, resulting in a dataset with 52 columns, each representing various attributes of the captured network traffic. A new categorical column 'Network' is added to label the data instances. Specifically, a '1' was assigned to instances originating from the BLE Mesh network captured through Bridgfy, while a '0' represented instances from conventional BLE devices. This comprehensive dataset, consisting of 98,515 rows, out of which 50814 samples belong to BLE mesh traffic, with 53 traffic features as shown in table 1.

Table 1. BT Mesh Network Traffic Features Extracted (Full Features)

1	No.	19	Delta time (end to start)	37	Reserved.5
2	Time	20	Access Address	38	Reserved.6
3	Source	21	Reserved.1	39	Tx Address
4	Destination	22	Length.1	40	Advertising Address
5	Protocol	23	Advertising Mode	41	Entry
6	Length	24	Reserved.2	42	IG bit
7	RSSI	25	Advertiser Set Identifier	43	Length.2
8	Channel Index	26	Offset units	44	Type
9	Flags	27	Aux PHY	45	UUID 16
10	Length of packet	28	CRC	46	Advertising Data
11	Header	29	CRC.1	47	Expert Info
12	Board	30	Reserved.3	48	Packet Header
13	Length of payload	31	Reserved.4	49	Undecoded
14	Protocol version	32	Address Resolved	50	Severity level
15	Packet counter	33	PHY	51	Group
16	Packet ID	34	Timestamp	52	Info
17	Reserved	35	Packet time (start to end)	53	Network
18	Event counter	36	PDU Type		52 + 1 (Labeled Column)

3.2 Data Preprocessing

In the process of constructing our model, we addressed missing data by employing techniques such as filling with mode values and utilizing specified strings. These steps were instrumental in ensuring that our dataset was complete and ready for analysis. Additionally, for attributes of object type, we applied encoding techniques including label encoding and one-hot encoding. These methods enabled us to represent categorical data in a numerical format, an essential step for effective model training.

Correlation measures the statistical relationship between two variables. In the context of feature reduction, we can use correlation to identify and remove highly correlated features, as they may carry redundant information. High correlation between two features implies that they provide similar information to the model. The steps in correlation feature reduction includes: (i) calculating the correlation matrix for all pairs of features, (ii) identify pairs of features with high correlation with a correlation threshold of 0.7, (iii) choose one feature from each highly correlated pair to keep and discard the others.

Algorithm 1. Label Encoding

```
Input: Categorical data feature with labels.
Output: Encoded feature with unique integers.
Initialize a dictionary to map labels to integers and an integer counter to start from 0.
For each unique label in the categorical feature:
        Check if the label is already in the dictionary.
        If(true):
                | do nothing.
        Else:
                | add the label to the dictionary and map it to the current integer counter.
        Increment the integer counter by 1.
End
Create a new feature or column to hold the encoded values.
Map the original labels to unique integers.
The encoded feature now contains unique integers for the original labels.
```

Algorithm 2. One-hot encoding

```
Input: Categorical data feature with labels.
Output: Binary columns for each category with 1s and 0s.
Determine the unique categories in the categorical feature.
For each data point in the original feature:
        Create a binary column for each unique category.
        If(true):
                | Assign 1 to the column corresponding to the data point's category.
        Else:
                | otherwise, set it to 0.
End
encoded feature now represents the categorical data with binary columns of 1s and 0s.
Output the dataset with the new encoded feature.
```

A subset of 26 columns was identified after correlation based feature reduction as shown in Table 2. The heat map of the correlation analysis is shown in Fig 2. This reduction in features aimed to streamline the dataset, focusing on variables that exhibited higher correlations and thereby optimizing the model's overall performance. This strategic feature selection process serves as a pivotal step in enhancing the relevance and predictive power of the dataset for subsequent analysis and modelling. This step was crucial in reducing data size, eliminating noise, and removing any unwanted data, all of which contributed to enhancing the overall efficiency and effectiveness of our machine learning model. In the domain of data preprocessing, an indispensable facet of our research, we embarked on a systematic journey designed to hone the quality and pertinence of our dataset. Our initial thrust centered on the meticulous pruning of columns containing superfluous or immaterial data. We exercised particular caution when examining attributes susceptible to time-related fluctuations, arising from the inherent variability of experimental conditions. This rigorous culling process resulted in the crafting of a refined dataset, thoughtfully curated to preserve only the most enlightening and steadfast features.

After this phase of data refinement, we delved into the application of data transformation techniques. These techniques were instrumental in enhancing the harmony and effectiveness of the remaining columns, serving to elevate data integrity and credibility. This preprocessing mission played an essential role in laying the groundwork for the ensuing stages of our research, laying the groundwork for the development and training of our machine learning models.

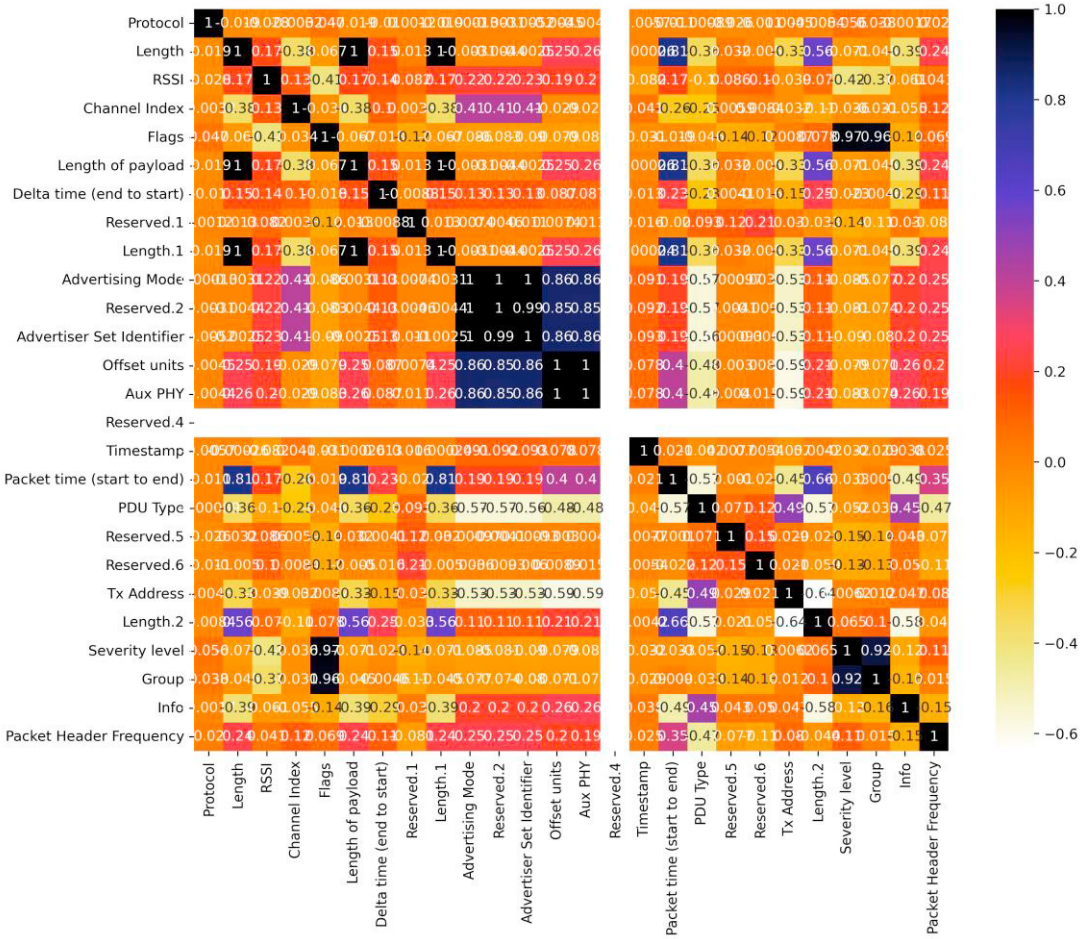


Fig 2. Heat map of correlation analysis

4. Experimentation and results

In the experimentation phase, we harnessed a laptop equipped with an Intel i5 processor, 8 GB RAM, a 512 GB SSD, and a potent 4 GB GTX 1650 Ti graphics card. Complementing our setup, we employed the nRF52840 dongle, a compact yet versatile USB dongle known for its extensive protocol support, including Bluetooth 5.4, Bluetooth mesh, Thread, Zigbee, 802.15.4, ANT, and 2.4 GHz proprietary protocols. This device, powered by the Nordic nRF52840 System-on-Chip (SoC) with 1 MB flash memory and 256 KB RAM, provided the foundation for our wireless device development and prototyping.

Table 2. Descriptive Overview of Dataset Columns after pre-processing.

1	Protocol	14	Aux PHY
2	Length	15	Reserved.4
3	RSSI	16	Timestamp
4	Channel Index	17	Packet time (start to end)
5	Flags	18	PDU Type
6	Length of payload	19	Reserved.5
7	Delta time (end to start)	20	Reserved.6
8	Reserved.1	21	Tx Address
9	Length.1	22	Length.2
10	Advertising Mode	23	Severity level
11	Reserved.2	24	Group
12	Advertiser Set Identifier	25	Info
13	Offset units	26	Packet Header Frequency

The configuration of the nRF52840 dongle was meticulous, featuring critical specifications such as the availability of various interfaces, LEDs, and buttons. To enable seamless development, we relied on nRF Connect for desktop, a versatile software tool that streamlined the configuration process. This entailed setting device parameters, including its name, serial number, operating mode, channel, power level, and security settings. The availability of Device Firmware Update (DFU) mode ensured programming flexibility, even in the face of bootloader issues. Furthermore, nRF Connect for desktop facilitated firmware updates, enhancing the versatility of the dongle. With the nRF52840 dongle primed and customized, we ventured into the realm of developing and prototyping wireless devices, leveraging its capabilities for our research and experimentation.

In the progression towards the construction of our machine learning model, we initiated the process by engaging in a preliminary data preparation endeavour within Jupyter Notebook. In our quest for model refinement, we harnessed correlation analysis to uncover related attributes among the remaining columns. This meticulous curation ultimately distilled the dataset down to a lean 26-column configuration, optimized for model training. The ensuing phase was characterized by comprehensive model evaluation, involving systematic experimentation with an array of machine learning algorithms. The basis for model selection was anchored in an exhaustive assessment of performance metrics, comprising accuracy, precision, recall, F1 score, specificity, balanced accuracy, Matthew's correlation coefficient (MCC), and the Receiver Operating Characteristic Area Under the Curve (ROC AUC). Through a rigorous evaluation of these performance indicators, we were able to discern and embrace the most effective model, a testament to our research and dedication. Fig. 3 compares the classifier outputs used in evaluating the dataset.

As we neared the crucial decision of selecting the best machine learning model, we created a comprehensive table. Table 3 contains a variety of machine learning models, each with its own unique set of parameters. It became our compass in the evaluation process, helping us systematically judge how each model performed with our dataset. We didn't just rely on one measure; instead, we considered several critical metrics like accuracy, precision, recall, F1 score, specificity (the true negative rate), balanced accuracy, Matthew's correlation coefficient (MCC), and the Receiver Operating Characteristic Area Under the Curve (ROC AUC).

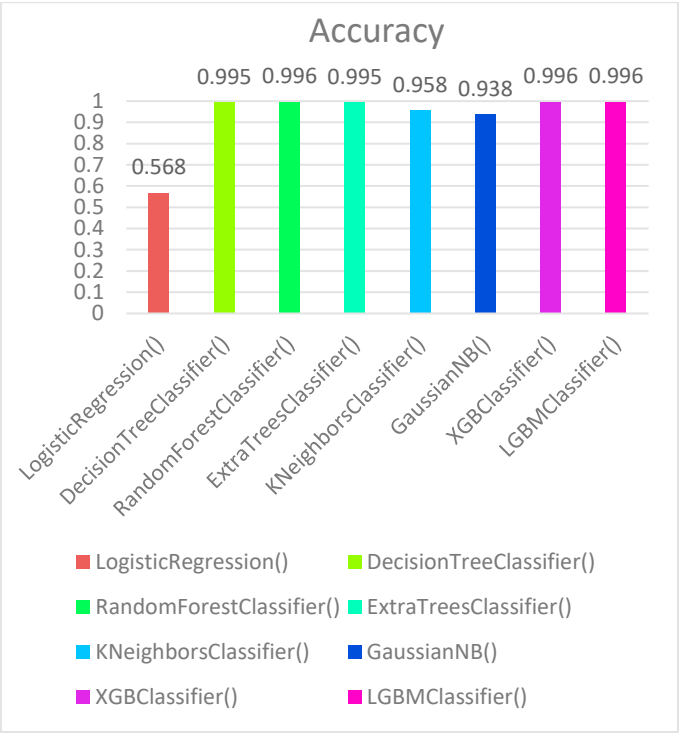


Fig 3. Comparison of Model Accuracies

From Table 3 it can be concluded that amongst the ML models tested XGB and LGBM classifiers have shown outstanding performance. Fig. 4 graphs the ROC curves. We conducted thorough testing for each parameter configuration and paid close attention to how well each model performed based on these metrics. This rigorous analysis gave us deep insights into the strengths and weaknesses of each model. It helped us make an informed choice when it came to picking the model that performed best. This approach, characterized by

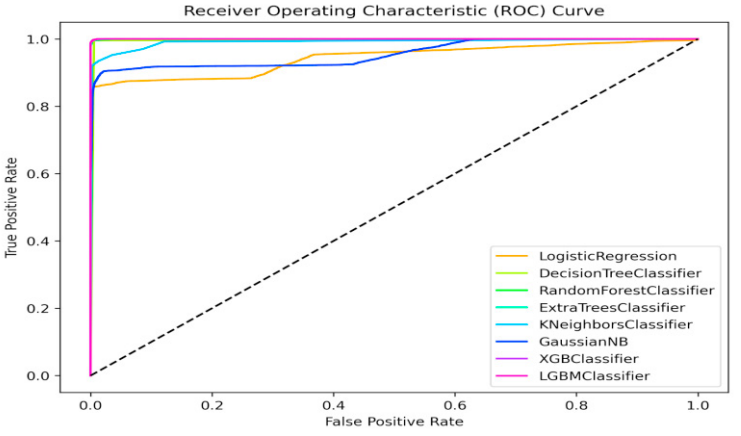


Fig 4. Receiver Operating Characteristic (ROC) Curves for Multiple Models

Table 3. Model Performance Metrics

Sl. No	Model Name	Accuracy	Precision	Recall	F1 Score	ROC_AUC_Score	Balanced Accuracy
1	LogisticRegression ()	0.568	0.546	0.99	0.704	0.552	0.552
2	DecisionTreeClassifier ()	0.995	0.995	0.996	0.995	0.995	0.996
3	RandomForestClassifier ()	0.996	0.997	0.995	0.996	0.996	0.996
4	ExtraTreesClassifier ()	0.995	0.996	0.995	0.996	0.995	0.996
5	KNeighborsClassifier ()	0.958	0.966	0.952	0.959	0.958	0.958
6	GaussianNB ()	0.938	0.984	0.894	0.937	0.939	0.941
7	**XGBClassifier ()**	**0.996**	**0.997**	**0.996**	**0.996**	**0.996**	**0.996**
8	**LGBMClassifier ()**	**0.996**	**0.996**	**0.997**	**0.997**	**0.996**	**0.996**

5. Conclusion

This paper proposed ML approach for detecting the very presence of BT mesh networks, as they have been misused by anti-social elements. The methods proposed can be used by the law enforcement agencies to detect any potential BT mesh networks in the surrounding. To address these concerns, we've introduced an innovative method for identifying mesh networking apps using packet analysis and machine learning. This method allows us to distinguish between traditional and mesh networking traffic, highlighting the power of data analysis and machine learning. Our research involved a series of steps, from setting up the mesh network with Bridgefy to capturing data using the nRF52840 Dongle and thorough data preparation. We carefully curated a dataset, selected essential features, and used machine learning models for detection. The results showcase the effectiveness of our approach, with specific models demonstrating impressive performance metrics. This paper develops a new dataset that can be used for BT mesh network detection. The dataset contains 52 BT mesh network traffic attributes. Filter based correlation feature analysis is employed further to optimize the feature size and reduced to 26 attributes with an accuracy of 99.6% tested with XGB and LGMB classifiers.

References

[1] K. Khaled, M A Hassan, M. Mohamed, A Madkour, and Sayed A. Nouh. (2023) “A review of security challenges and solutions in wireless sensor networks.” *Journal of Al-Azhar University Engineering Sector* **18(69)**: 914-938.

[2] I. F. Akyildiz, X. Wang, and W. Wang. (2005) “Wireless mesh networks: a survey.” *Computer Networks* **47(4)**: 445-487.

[3] M. R. Ghori, T.-C. Wan, and G. C. Sodhy. (2020) “Bluetooth Low Energy Mesh Networks: Survey of Communication and Security Protocols.” *Sensors* **20(12)**: 1-35.

[4] M. Sugita, H. Matsue, K. Yamaguchi, T. Shimodaira, M. Shirotori, and S. Nanamatsu. (2015) “Development strategy of disaster-resistant WiFi mesh networks and the disaster monitoring and reporting system in a local area.” *Proceeding of the 2015 International Conference on Information Networking (ICOIN),* Cambodia.

[5] W. Hirushan, S. Shamilka, R. Hirshan, and M. Haleem. (2023) “Mesh Network Based Mobile Communication.” *Proceeding of the IEEE International Conference on Advanced Networks and Telecommunications Systems (IEEE ANTS)*, Gandhinagar.

[6] B. Zhang, Y. Wang, L. Wei, Q. Jin, and A. V. Vasilakos. (2018) “BLE mesh: a practical mesh networking development framework for public safety communications.” *Tsinghua Science and Technology* **23(3)**: 1-13.

[7] Pierluigi Locatelli, M. Perri, D. M. J. Gutierrez, A. Andrea Lacava, and Francesca Cuomo. (2023) “Device discovery and tracing in the Bluetooth Low Energy domain.” *Computer Communications* **202**: 42-56.

[8] Z. Guo, I. G. Harris, Y. Jiang, and Lih-feng Tsaur. (2017) “An efficient approach to prevent Battery Exhaustion Attack on BLE-based mesh networks.” *Proceedings of the 2017 International Conference on Computing, Networking and Communications (ICNC), Silicon Valley, CA, USA*.

[9] Hao Yang, Haiyun Luo, Fan Ye, Songwu Lu, and Lixia Zhang. (2004) “Security in mobile ad hoc networks: challenges and solutions.” *IEEE Wireless Communications* **11(1)**: 38 - 47.

[10] K. Saxena. (2023) "That's a 'Sign'!: How Placards Communicated Disapproval of the Citizenship Amendment Act (CAA) in India." *Communication Studies*, **74(4)**: 322-336.

[11] Albrecht M. R., Blasco J., Jensen R. B., and Mareková L. (2021) "Mesh Messaging in Large-Scale Protests: Breaking Bridgefy." *in Topics in Cryptology – CT-RSA 2021, Lecture Notes in Computer Science* **12704**: 375–398.

[12] Sadaf Masood, Yousaf Saeed, Harun Jamil, Nagwan M Abdel Samee, Hayam Alamro, Mohammed Saleh Ali Muthanna, and Abdukodir Khakimov. (2023) "Detecting and Preventing False Nodes and Messages in Vehicular Ad-Hoc Networking (VANET)." *IEEE Access* **11**: 93920-93934.

[13] F. S. Al-Anzi. (2022) "Design and analysis of intrusion detection systems for wireless." *Digital Communications and Networks* **8(6)**: 1068-1076.

[14] S. Karunaratne, and H. Gacanin. (2019) "An Overview of Machine Learning Approaches in Wireless Mesh Networks," *IEEE Communications Magazine* **57(4)**: 102 - 108.

[15] G. Alamer, S. Alyahya, and H. Al-Dossari. (2023) "Identifying Users and Developers of Mobile Apps in Social Network Crowd." *Electronics* **12(16)**: 1-17.